

Geometry



Area and perimeter

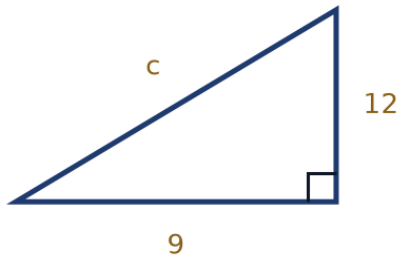
- 1 A rectangle has the dimensions shown below. Find its area and perimeter.



- 2 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.
- 3 A triangle has base 15 and height 10. Find its area.

The Pythagorean theorem

- 4 A right triangle has the two legs shown below. Find the length of the hypotenuse C .



- 5 A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

Volume and surface area

- 6 Find the volume of a cylinder with radius 4 and height 9, in terms of π .
- 7 Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

Similar triangles and proportional reasoning

- 8 Triangle ABC is similar to triangle DEF , with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF .
- 9 A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.
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Angle relationships

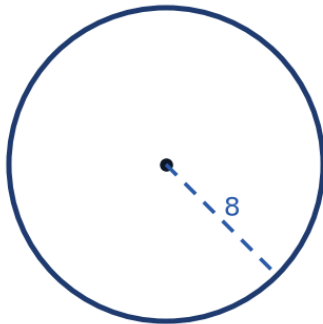
- 10 Two parallel lines are cut by a transversal. If one angle measures 72° , find its corresponding angle and its co-interior (same-side interior) angle.
- 11 The three angles of a triangle are in the ratio $2 : 3 : 4$. Find each angle.
- 12 Two angles are complementary, and one is 34° more than the other. Find both angles.
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Coordinate geometry: distance and midpoint

- 13 Find the distance between the points $(3, -2)$ and $(9, 6)$.
- 14 Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.
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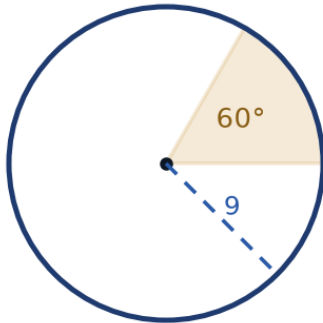
Circles: area, circumference, and equation

- 15 A circle has the radius shown below. Find its area and circumference, in terms of π .



- 16 Write the equation of a circle with center $(3, -4)$ and radius 6.

- 17 A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π .



Composite and shaded regions

- 18 A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π .
- 19 A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

A well-designed word problem: designing a park

- 20 This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π . (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π .

Multiple choice — easy

- 21 A rectangle has length 23 and width 5. Find its area and perimeter.
- a. 115, 56
 - b. 56, 115
 - c. 119, 56
 - d. 115, 54
- 22 A rectangle has length 5 and width 23. Find its area and perimeter.
- a. 115, 56
 - b. 56, 115
 - c. 119, 56
 - d. 115, 54
- 23 A rectangle has length 17 and width 15. Find its area and perimeter.
- a. 255, 64
 - b. 64, 255
 - c. 259, 64
 - d. 255, 62
- 24 A rectangle has length 30 and width 20. Find its area and perimeter.
- a. 100, 600
 - b. 604, 100
 - c. 600, 98
 - d. 600, 100

- 25 A rectangle has length 27 and width 6. Find its area and perimeter.
- a. 162, 66
 - b. 66, 162
 - c. 166, 66
 - d. 162, 64
- 26 A rectangle has length 9 and width 6. Find its area and perimeter.
- a. 54, 28
 - b. 58, 30
 - c. 54, 30
 - d. 30, 54
- 27 A rectangle has length 4 and width 13. Find its area and perimeter.
- a. 52, 34
 - b. 34, 52
 - c. 52, 32
 - d. 56, 34
- 28 A rectangle has length 30 and width 18. Find its area and perimeter.
- a. 540, 94
 - b. 96, 540
 - c. 544, 96
 - d. 540, 96

- 29 A rectangle has length 30 and width 24. Find its area and perimeter.
- a. 720, 106
 - b. 720, 108
 - c. 108, 720
 - d. 724, 108
- 30 A rectangle has length 26 and width 16. Find its area and perimeter.
- a. 416, 84
 - b. 416, 82
 - c. 420, 84
 - d. 84, 416
- 31 A rectangle has length 23 and width 3. Find its area and perimeter.
- a. 69, 50
 - b. 52, 69
 - c. 69, 52
 - d. 73, 52
- 32 A rectangle has length 20 and width 15. Find its area and perimeter.
- a. 304, 70
 - b. 70, 300
 - c. 300, 68
 - d. 300, 70

33 A rectangle has length 14 and width 25. Find its area and perimeter.

- a. 354, 78
- b. 350, 76
- c. 350, 78
- d. 78, 350

34 A rectangle has length 13 and width 23. Find its area and perimeter.

- a. 72, 299
- b. 303, 72
- c. 299, 70
- d. 299, 72

35 A rectangle has length 23 and width 8. Find its area and perimeter.

- a. 184, 60
- b. 62, 184
- c. 184, 62
- d. 188, 62

Multiple choice — medium

36 A right triangle has legs 3 and 4. Find the length of the hypotenuse c .

- a. 5
- b. 7
- c. 3
- d. 9

- 37 A right triangle has legs 60 and 63. Find the length of the hypotenuse c .
- a. 85
 - b. 89
 - c. 87
 - d. 123
- 38 A right triangle has legs 8 and 15. Find the length of the hypotenuse c .
- a. 17
 - b. 19
 - c. 23
 - d. 15
- 39 A right triangle has legs 7 and 24. Find the length of the hypotenuse c .
- a. 27
 - b. 25
 - c. 31
 - d. 23
- 40 A right triangle has legs 9 and 40. Find the length of the hypotenuse c .
- a. 49
 - b. 39
 - c. 41
 - d. 43

- 41 A right triangle has legs 20 and 21. Find the length of the hypotenuse c .
- a. 27
 - b. 31
 - c. 41
 - d. 29
- 42 A right triangle has legs 24 and 45. Find the length of the hypotenuse c .
- a. 53
 - b. 51
 - c. 69
 - d. 49
- 43 A right triangle has legs 11 and 60. Find the length of the hypotenuse c .
- a. 71
 - b. 63
 - c. 61
 - d. 59
- 44 A right triangle has legs 24 and 70. Find the length of the hypotenuse c .
- a. 72
 - b. 74
 - c. 94
 - d. 76

- 45 A right triangle has legs 15 and 36. Find the length of the hypotenuse c .
- a. 51
 - b. 37
 - c. 39
 - d. 41
- 46 A right triangle has legs 18 and 80. Find the length of the hypotenuse c .
- a. 98
 - b. 84
 - c. 82
 - d. 80
- 47 A right triangle has legs 40 and 42. Find the length of the hypotenuse c .
- a. 58
 - b. 56
 - c. 60
 - d. 82
- 48 A right triangle has legs 27 and 120. Find the length of the hypotenuse c .
- a. 125
 - b. 123
 - c. 121
 - d. 147

- 49 A right triangle has legs 12 and 35. Find the length of the hypotenuse c .
- a. 35
 - b. 37
 - c. 39
 - d. 47
- 50 A right triangle has legs 9 and 12. Find the length of the hypotenuse c .
- a. 13
 - b. 21
 - c. 15
 - d. 17
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Multiple choice — hard

- 51 Find the volume of a cylinder with radius 7 and height 13, in terms of π .
- a. 2548π
 - b. 643π
 - c. 91π
 - d. 637π
- 52 Find the volume of a cylinder with radius 6 and height 4, in terms of π .
- a. 576π
 - b. 24π
 - c. 150π
 - d. 144π

- 53 Find the volume of a cylinder with radius 5 and height 13, in terms of π .
- a. 65π
 - b. 1300π
 - c. 325π
 - d. 331π
- 54 Find the volume of a cylinder with radius 3 and height 6, in terms of π .
- a. 216π
 - b. 18π
 - c. 60π
 - d. 54π
- 55 Find the volume of a cylinder with radius 9 and height 7, in terms of π .
- a. 2268π
 - b. 63π
 - c. 573π
 - d. 567π
- 56 Find the distance between the points $(-2, -7)$ and $(-14, -2)$.
- a. 11
 - b. 17
 - c. 15
 - d. 13

57 Find the distance between the points $(-10, 2)$ and $(-15, -10)$.

a. 13

b. 11

c. 15

d. 17

58 Find the distance between the points $(5, -8)$ and $(29, -15)$.

a. 23

b. 27

c. 31

d. 25

59 Find the distance between the points $(-5, -9)$ and $(-14, 31)$.

a. 41

b. 43

c. 39

d. 49

60 Find the distance between the points $(5, 9)$ and $(-9, 57)$.

a. 50

b. 52

c. 62

d. 48

- 61 A square with side length 8 has a circle of radius 2 removed from its center. Find the remaining area, in terms of π .
- a. 64π
 - b. $64 - 4\pi$
 - c. $68 - 4\pi$
 - d. $64 - 6\pi$
- 62 A square with side length 19 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π .
- a. $365 - 25\pi$
 - b. $361 - 25\pi$
 - c. $361 - 27\pi$
 - d. 361π
- 63 A square with side length 12 has a circle of radius 2 removed from its center. Find the remaining area, in terms of π .
- a. 144π
 - b. $144 - 4\pi$
 - c. $148 - 4\pi$
 - d. $144 - 6\pi$
- 64 A square with side length 16 has a circle of radius 4 removed from its center. Find the remaining area, in terms of π .
- a. $260 - 16\pi$
 - b. 256π
 - c. $256 - 18\pi$
 - d. $256 - 16\pi$

- 65 A square with side length 8 has a circle of radius 3 removed from its center. Find the remaining area, in terms of π .
- a. $64 - 9\pi$
 - b. $68 - 9\pi$
 - c. 64π
 - d. $64 - 11\pi$